

POCKETSMART AI

Your Smart Budget & Recommendation Assistant

Document	06 – Project Testing Report
Project	PocketSmart AI
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Phase	Phase 6 – Testing

1. Testing Objective

The objective of testing is to verify that the major PocketSmart AI functions work correctly with valid user inputs and that generated recommendations, database storage, history, navigation, and shopping search links behave as expected.

2. Testing Approach

The application was tested by running the FastAPI web application and entering representative real-world budget inputs in the available planner modules. The observed results were compared with the expected application behavior.

3. Test Case Report

TC ID	Test Case	Input / Action	Expected Result	Actual Result	Status
TC0 1	User Registration	Enter valid registration details	User account should be created	Registration completed successfully	PASS
TC0 2	User Login	Enter registered credentials	User should be logged in	Login completed successfully	PASS
TC0 3	Home Planner	Budget: ■50,000 with home planning inputs	System should generate a home budget recommendation	Home recommendation generated successfully	PASS
TC0 4	Party Planner	■50,000; Birthday; 50 guests; Hall; Blue; Vegetarian; Simple	System should generate a party recommendation	Party recommendation generated successfully	PASS
TC0 5	Jewelry Planner	■25,000; Ring; Wedding; Gold; Simple; Women	System should generate a jewelry recommendation	Jewelry recommendation generated successfully	PASS
TC0 6	Recommendation Generation	Submit valid planner data	Recommendation should be returned	Recommendation returned successfully	PASS
TC0 7	Fallback Recommendation	Generate recommendation when Gemini service is not configured	System should provide fallback recommendation	Rule-based fallback recommendation generated	PASS

TC ID	Test Case	Input / Action	Expected Result	Actual Result	Status
TC08	Shopping Links	Open generated shopping search links	Relevant search links should be displayed	Shopping links worked successfully	PASS
TC09	Recommendation History	Save recommendation and open History	Saved recommendation should appear in history	History displayed saved recommendations	PASS
TC10	Logout	Click Logout	User session should end	Logout completed successfully	PASS

4. Real-World Budget Testing

The application was tested with representative budget values across the available domains. A \$50,000 budget was used for Home Planner and Party Planner testing, while a \$25,000 budget was used for Jewelry Planner testing. The generated recommendations were displayed successfully.

5. Testing of Recommendation Logic

The recommendation flow was verified using the configured recommendation service. When Gemini AI is available, the application can use the AI service; when it is not configured, the application uses a rule-based fallback so that the planner can still return a recommendation.

6. Testing of Database and History

Recommendation results were stored and retrieved through the application's database-backed history functionality. The History page was checked to confirm that saved recommendations could be viewed.

7. Test Result Summary

Item	Result
Total Test Cases	10
Passed	10
Failed	0
Overall Test Status	All tested cases passed

8. Bug Fix / Verification Process

During development, issues were verified by repeating the affected workflow after corrections. The final verification focused on planner submission, recommendation generation, shopping links, history storage, navigation, and authentication behavior.

9. Testing Conclusion

The tested PocketSmart AI workflows operated as expected for the selected test inputs. The major planner modules, recommendation generation, fallback behavior, shopping links, recommendation history, and authentication flow were successfully verified.

Note: This report records the test scenarios and results performed for the project. Additional testing can be performed with more input combinations before final deployment.